

Homework 12.

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1. $\{X_n\}$: sequence of random variables.

a) $\lim_{n \rightarrow \infty} E[|X_n|] = 0$

$$\lim_{n \rightarrow \infty} P(|X_n| > \varepsilon) = 0 \quad \forall \varepsilon > 0.$$

from Markov inequality

$$P(|X_n| > \varepsilon) \leq \frac{E[|X_n|]}{\varepsilon}$$

$$\therefore \lim_{n \rightarrow \infty} P(|X_n| > \varepsilon) \leq \lim_{n \rightarrow \infty} \frac{E[|X_n|]}{\varepsilon} = 0 \quad \text{Q.E.D.}$$

b) $X_n: \Omega \rightarrow \{0, 1\}$, $P(X_n = 1) = 1/n$
 $\Rightarrow P(X_n = 0) = 1 - P(X_n = 1) = 1 - \frac{1}{n} = \frac{n-1}{n}$

$$\lim_{n \rightarrow \infty} X_n(\omega) = 0, \quad \forall \omega \in A \in \mathcal{F}, \quad P(A) = 1$$

$$P(\{\omega \mid X_n(\omega) \rightarrow 0\}) = 1$$

i. $X_n \xrightarrow{\text{a.s.}} 0$

$$X_n(\omega) = \begin{cases} 1 & \omega \in [0, 1/n] \\ 0 & \omega \in (1/n, 1] \end{cases}$$

ii) $X_n \xrightarrow{\text{a.s.}} 0$

if X_n are independent

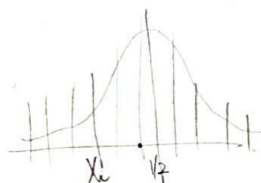
$$\lim_{n \rightarrow \infty} \prod_{i=1}^n X_i = 0$$

2. $X_i \sim \text{Unif}(0, 1)$

Y_n : median of $X_1, X_2, \dots, X_{n+1}$

$$X_1 \leq X_2 \leq \dots \leq X_{n+1}$$

$$Y_n = X_{n+1} \quad \text{show } Y_n \xrightarrow{P} 1/2$$



$$3. X_1, X_2, \dots \sim N(0, 1^2) \quad \phi_n(t) = e^{-t^2/2}$$

$$Y_n = \sum_{i=1}^n \frac{X_i}{2^i} = \frac{1}{2}X_1 + \frac{1}{4}X_2 + \frac{1}{8}X_3 + \dots + \frac{1}{2^n}X_n$$

$$M_{aX+bY}(t) = E[e^{t(aX+bY)}] = E[e^{taX}]E[e^{tbY}] \\ = M_X(at)M_Y(bt)$$

$$\therefore M_{Y_n}(t) = M_{X_1}(\frac{1}{2}t)M_{X_2}(\frac{1}{4}t)\dots M_{X_n}(\frac{1}{2^n}t)$$

$$\Rightarrow \phi_{Y_n}(t) = \prod_{i=1}^n \phi_{X_i}(\frac{t}{2^i}) = \prod_{i=1}^n e^{-(t/2^i)^2/2} = \prod_{i=1}^n e^{-\frac{t^2}{4^i \cdot 2}}$$

$$= \exp\left[-\sum_{i=1}^n \frac{t^2}{2 \cdot 4^i}\right] = \exp\left[-\frac{t^2}{2} \cdot \sum_{i=1}^n \frac{1}{4^i}\right]$$

$$\lim_{n \rightarrow \infty} \phi_{Y_n}(t) = \lim_{n \rightarrow \infty} \exp\left[-\frac{t^2}{2} \sum_{i=1}^n \frac{1}{4^i}\right] = \exp\left[-\frac{t^2}{2} \cdot \frac{1}{1-\frac{1}{4}}\right] = \exp\left[-\frac{t^2}{6}\right]$$

$$\therefore \lim_{n \rightarrow \infty} Y_n \sim N(0, \frac{1}{3})$$

$$4. X_1, X_2, \dots \sim \text{Unif}(-0.7, 0.7)$$

$$Y = X_1 + X_2 + \dots + X_{100}$$

$$Y \sim N(\mu, \sigma^2)$$

$$P(Y > 3) + P(Y < -3)$$

$$\mu = E[X_1 + \dots + X_{100}] = 0$$

$$\sigma^2 = V[X_1 + \dots + X_{100}] = 100^2 V[X_1] = 100 \cdot \frac{(1.4)^2}{12} = \frac{196}{12} = \frac{49}{3}$$

$$\therefore Y \sim N(0, \frac{49}{3})$$

$$P(Y > 3) + P(Y < -3) = 2P(Y > 3) = 2\left(1 - \Phi\left(\frac{3}{\sqrt{49/3}}\right)\right) =$$

$$2 - 2\Phi\left(\frac{\sqrt{3}}{7}\right)$$